

MLCV Home Assignment - 01

Performance Evaluation of Deep Learning Optimization and Regularization Techniques for Multi-Class Product Classification

Background

The **Otto Group** is one of the world's largest e-commerce companies, operating in more than 20 countries with millions of products listed across different online platforms. Due to variations in regional cataloging systems, identical products are often assigned different product categories. This inconsistency affects inventory management, recommendation systems, business analytics, and customer experience.

To address this challenge, Otto Group has released a dataset consisting of **93 numerical product features** and corresponding product category labels. Your objective is to develop a robust Deep Neural Network (DNN) model capable of accurately classifying products into the correct category while minimizing overfitting and improving generalization.

As a Deep Learning Engineer, you are required to investigate how different optimization algorithms and regularization techniques influence the convergence, accuracy, and generalization ability of the model.

Problem Statement

Develop a Deep Neural Network (DNN) to classify products into one of **nine product categories** using the provided tabular dataset. Starting with a baseline model, systematically investigate the impact of different **optimizers**, **regularization techniques**, **dropout rates**, and **early stopping** on the model's performance.

Based on experimental results, recommend the most effective training strategy that achieves high classification accuracy while maintaining good generalization.

Dataset

Dataset: Otto Group Product Classification Challenge

<https://www.kaggle.com/competitions/otto-group-product-classification-challenge/data>

Input: `train.csv`

- Samples: **61,878**
- Features: **93 numerical features**
- Output Classes: **9 product categories**

Visualization (10 Marks)

Generate and interpret the following plots:

1. Training Accuracy vs Epochs
2. Validation Accuracy vs Epochs
3. Training Loss vs Epochs
4. Validation Loss vs Epochs
5. Optimizer Comparison (Bar Chart)
6. Regularization Comparison
7. Dropout Comparison
8. Confusion Matrix
9. Classification Report

Part A: Data Preprocessing (10 Marks)

Perform the following preprocessing steps:

1. Load the dataset.
2. Check for missing values.
3. Encode target labels.
4. Normalize or standardize the input features.
5. Split the dataset into:
 - Training Set
 - Validation Set
 - Test Set
6. Justify the preprocessing techniques used.

Part B: Baseline Deep Neural Network (10 Marks)

Design a baseline MLP with the following specifications:

- Input Layer: 93 neurons
- Hidden Layer 1: 256 neurons
- Hidden Layer 2: 128 neurons
- Hidden Layer 3: 64 neurons
- Hidden Layer 4: 32 neurons
- Output Layer: 9 neurons (Softmax)

Compile the model using:

- Optimizer: SGD
- Learning Rate: 0.01
- Loss Function: Categorical Crossentropy

Train the model for **200 epochs** without:

- Regularization
- Dropout
- Early Stopping

Record:

- Training Accuracy
- Validation Accuracy
- Training Loss
- Validation Loss

Part C: Optimizer Comparison (20 Marks)

Using the same architecture, train the model with the following optimizers:

- SGD
- SGD with Momentum

- RMSProp
- Adam
- AdamW

For each optimizer, compare:

- Convergence speed
- Final training accuracy
- Validation accuracy
- Test accuracy
- Number of epochs required for convergence
- Training stability

Present the results in a comparative table.

Part D: Effect of Regularization (15 Marks)

Using the best-performing optimizer from Part C, investigate the effect of regularization.

Train the model using:

- No Regularization
- L1 Regularization
- L2 Regularization
- L1 + L2 Regularization

Compare:

- Training accuracy
- Validation accuracy
- Test accuracy
- Generalization gap
- Loss curves

Comment on how regularization affects model complexity and overfitting.

Part E: Effect of Dropout (15 Marks)

Using the best optimizer and regularization method, investigate the effect of Dropout.

Train the model using the following dropout rates:

- 0.0
- 0.2
- 0.4
- 0.5

Analyze:

- Training accuracy
- Validation accuracy
- Test accuracy
- Generalization gap

Determine the optimal dropout rate.

Part F: Early Stopping (10 Marks)

Train the final model:

Model A

Without Early Stopping

Model B

With Early Stopping

Use:

- Patience = 10
- Monitor = Validation Loss

Compare:

- Total training epochs
- Training time
- Best validation accuracy
- Test accuracy

Discuss the advantages of Early Stopping.

Part G: Comparative Analysis (10 Marks)

Prepare a summary table.

Model	Optimizer	Regularization	Dropout	Early Stopping	Validation Accuracy	Test Accuracy

Discussion Questions (20 Marks)

Type answers for all the following questions based on your experimental observations:

1. Which optimizer converged the fastest? Provide evidence using the learning curves.
2. Why did some optimizers achieve better validation accuracy than others?
3. Did the model exhibit overfitting? How did you identify it?
4. Compare the effects of L1 and L2 regularization on the model's performance.
5. Why did increasing the dropout rate beyond a certain value reduce the model's performance?
6. How did Early Stopping affect the total training time and model generalization?
7. Which combination of optimizer, regularization, dropout, and early stopping produced the best model? Justify your answer using quantitative results.
8. Suppose the Otto Group doubles the size of its product catalog. Which optimization strategy would you recommend for faster convergence and better scalability? Explain your reasoning.

Note: Only Jupyter codebook with all results with your details must be uploaded by 7th August 2026